A2	PROSOIL	BOREHOLE NO: BH_BRRL_N-13	Longitude: E 91°6'16.45"	Latitude: N 22°56'34.37"
		Name of Project: Engineering Soil mapping of Noakhali & Duptila as per soil classification of Road Master Plan Noakhali, Feni, Bhola, Chittagong, Dohazari & Cox's Bazar.		
CLIENT : BRRL.		Type of Drill Rig: Mechanical Rotary	Borehole Depth: 51.0m	RL: 3.5m below from Road level.
Location: Laxminagar Pur, Sotuvanga, Begumgonj, Noakhali.		Commencement Date: 06-06-2016	Completed Date: 10-06-2016	G. W. T : 2.39m Below from E. G. L.

THICKNESS	SOIL STRATA	BORE LOG	SAMPLING From (m)	SAMPLING To (m)	SPT-N Value Blow/300mm N1=150 (mm)	N2=300 (mm)	N3=450 (mm)	N=N2+N3	DISTURBED SAMPLE	DEPTH (m) SPT VALUES (BLOWS / 300mm)	% Sand	% Passing 200	% LL	% PI	SP Gravity	% W_n
13.5m	Gray 10YR (6/1), very loose to very dense, silty sand, SM.		1.5	1.95	0	0	2	2	UD-1	2						
									D-1							
			3	3.45	1	1	2	3	UD-2	3						
									D-2							
			4.5	4.95	1	0	5	5	UD-3	5						
									D-3							
			6	6.45	1	2	16	18	D-4	18						
			7.5	7.95	5	8	15	23	UD-4	23						
									D-5							
			9	9.45	3	2	19	21	UD-5	21						
									D-6							
			10.5	10.95	0	10	29	39	D-7	39						
			12	12.45	6	4	31	35	D-8	35						
			13.5	13.95	3	6	31	37	D-9	37						
18.0m	Gray 10YR (5/1), very dense, silty sand, SM.		15	15.45	4	11	32	43	UD-6	43						
									D-10							
			16.5	16.95	1	5	41	46	UD-7	46						
									D-11							
			18	18.45	0	17	33	50	D-12	50						
			19.5	19.95	10	7	43	50	UD-8	50						
									D-13							
			21	21.45	7	14	36	50	UD-9	50						
									D-14							
			22.5	22.95	8	6	44	50	D-15	50						
			24	24.45	1	5	45	50	UD-10	50						
									D-16							
			25.5	25.95	4	19	31	50	D-17	50						
			27	27.45	6	10	40	50	UD-11	50						
									D-18							
			28.5	28.95	3	19	31	50	D-19	50						

RL WITH REPECT TO SUB.

LEGEND FOR BORELOG AND GRAIN SIZE DISTRIBUTION (BASED ON 'ASTM CLASSIFICATION D-2487')

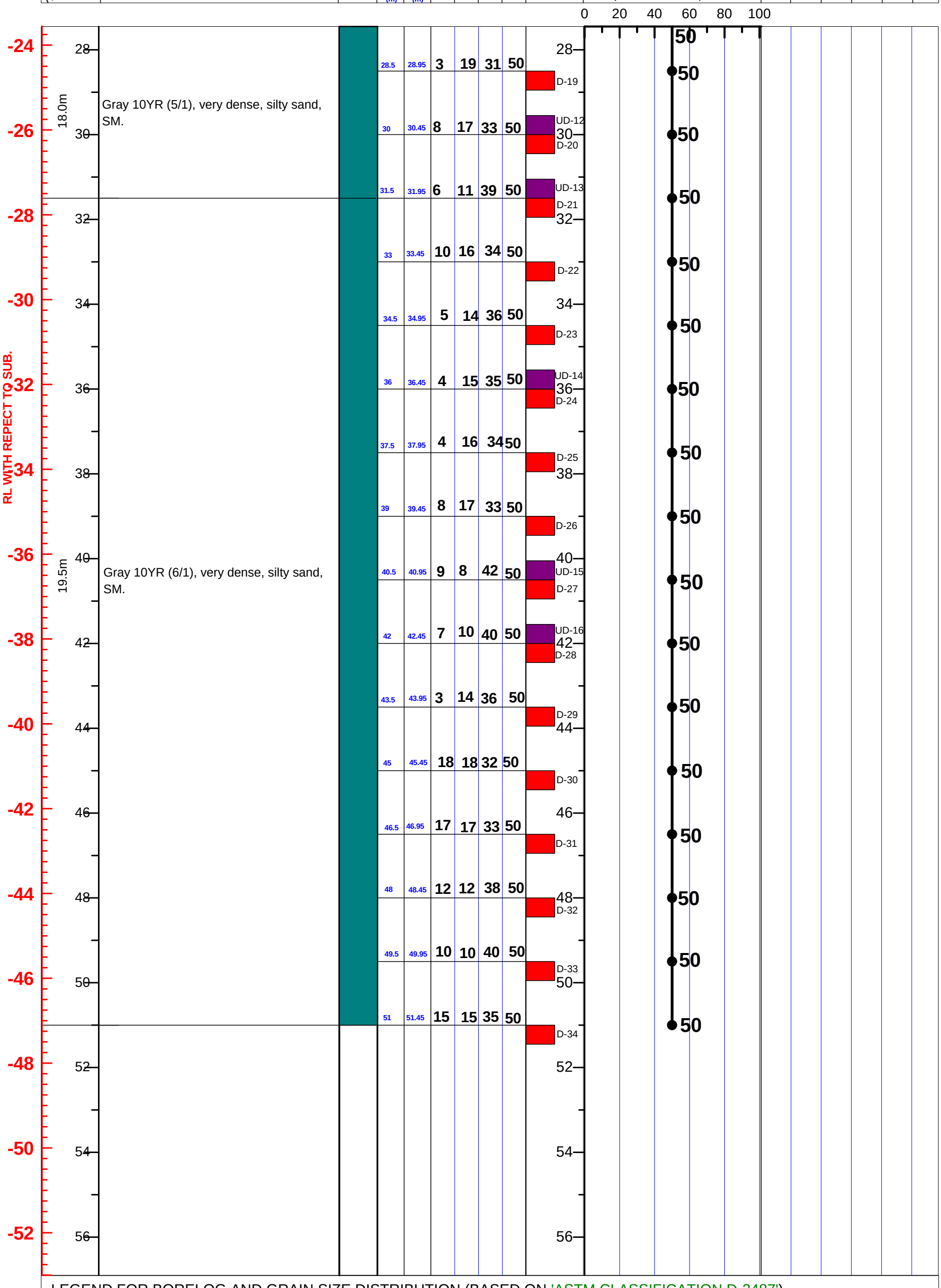
PLASTIC CLAY | SILT | PEAT/HIGHLY ORGANIC SOIL | UNDISTURBED SAMPLE

MICACEOUS FINE SAND/SILT | SAND | SHALE | DISTURBED SAMPLE

N- BLOWS DELIVERED PER FOOT BY A 140 LB HAMMER FALLING 30 INCHES

MAX^{m} PENETRATION= 50 BLOWS. AFTER 50 BLOWS PENETRATION DEPTH WERE MEASURED 50/X = BLOWS 50, "X" MM PENETRATION left of 300 mm

01	BRRL PROSOIL	BOREHOLE NO: BH_BRRL_N-13	Longitude: E 91°58'42.17"	Latitude: N 22°56'34.37"
		Name of Project: Engineering Soil mapping of Noakhali & Duptila as per soil classification of Road Master Plan Noakhali, Feni, Bhola, Chittagong, Dohazari & Cox's Bazar.		
CLIENT : BRRL.		Type of Drill Rig: Mechanical Rotary	Borehole Depth: 51.0m	RL: 3.5m below from Road level.
Location: Laxminagar Pur, Sotuvanga, Begumgonj, Noakhali.		Commencement Date: 06-06-2016	Completed Date: 10-06-2016	G. W. T : 2.39m Below from E. G. L.


THICKNESS
SOIL STRATA
BORE LOG
SAMPLING
From (m)
To (m)
SPT-N Value Blow/300mm
N1=150 (mm)
N2=300 (mm)
N3=450 (mm)
N=N2+N3
DISTURBED SAMPLE
DEPTH (m) SPT VALUES (BLOWS / 300mm)
% Sand
% Passing 200
% LL
% PI
SP Gravity
% Wn
0 20 40 60 80 100
RL WITH REPECT TO SUB.
-24 -26 -28 -30 -32 -34 -36 -38 -40 -42 -44 -46 -48 -50 -52
18.0m
Gray 10YR (5/1), very dense, silty sand, SM.
19.5m
Gray 10YR (6/1), very dense, silty sand, SM.
28.5 28.95 3 19 31 50 D-19
30 30.45 8 17 33 50 UD-12
30.95 31.45 D-20
31.5 31.95 6 11 39 50 UD-13
32.45 32.95 D-21
33 33.45 10 16 34 50 D-22
33.95 34.45
34.5 34.95 5 14 36 50 D-23
35.45 35.95
36 36.45 4 15 35 50 UD-14
36.95 37.45 D-24
37.5 37.95 4 16 34 50
38.45 38.95 D-25
39 39.45 8 17 33 50
39.95 40.45 D-26
40.5 40.95 9 8 42 50 UD-15
41.45 41.95 D-27
42 42.45 7 10 40 50 UD-16
42.95 43.45 D-28
43.5 43.95 3 14 36 50
44.45 44.95 D-29
45 45.45 18 18 32 50
45.95 46.45 D-30
46.5 46.95 17 17 33 50
47.45 47.95 D-31
48 48.45 12 12 38 50
48.95 49.45 D-32
49.5 49.95 10 10 40 50
50.45 50.95 D-33
51 51.45 15 15 35 50
51.95 52.45 D-34
50 50 50

A2	PROSOIL	BOREHOLE NO: BH_BRRL_N_14	Longitude: E 91°6'31.47	Latitude: N 23°3'7.42
		Name of Project: Engineering Soil mapping of Noakhali & Duptila as per soil classification of Road Master Plan Noakhali, Feni, Bhola, Chittagong, Dohazari & Cox's Bazar		
CLIENT : BRRL.		Type of Drill Rig: Mechanical Rotary	Borehole Depth: 51.0m	RL: 2.12m below from Road Level.
Location: Vodogaon, Amisha para, Sonaimuri, Noakhali.		Commencement Date: 06-06-2016	Completed Date: 10-06-2016	G. W. T : 1.83m Below from E. G. L.

THICKNESS	SOIL STRATA	BORE LOG	SAMPLING From (m)	SAMPLING To (m)	SPT-N Value Blow/300mm N1=150 (mm)	N2=300 (mm)	N3=450 (mm)	N=N2+N3	DISTURBED SAMPLE	DEPTH (m) SPT VALUES (BLOWS / 300mm)	% Sand	% Passing 200	% LL	% PI	SP Gravity	% W_n
21.0m	Brown 7.5YR (5/4), soft to very stiff, Low Plasticity clay with sand, CL.		1.5	1.95	0	1	2	3	UD-1	3						
									D-1							
			3	3.45	1	1	5	6	UD-2	6						
									D-2							
			4.5	4.95	1	1	7	8	UD-3	8						
									D-3							
			6	6.45	1	2	7	9	UD-4	9						
									D-4							
			7.5	7.95	2	3	7	10	UD-5	10						
									D-5							
			9	9.45	2	1	8	9	UD-6	9						
									D-6							
			10.5	10.95	2	5	11	16	D-7	16						
			12	12.45	1	6	9	15	UD-7	15						
									D-8							
			13.5	13.95	1	5	12	17	D-9	17						
			15	15.45	3	3	12	15	UD-8	15						
									D-10							
			16.5	16.95	1	3	12	15	D-11	15						
			18	18.45	3	4	14	18	D-12	18						
			19.5	19.95	3	2	13	15	UD-9	15						
									D-13							
			21	21.45	4	4	16	20	UD-10	20						
									D-14							
21.0m	Gray 7.5YR (5/1), dense to very dense, Low Plasticity Silt with Sand, ML		22.5	22.95	2	5	14	19	D-15	19						
			24	24.45	3	5	17	22	UD-11	22						
									D-16							
			25.5	25.95	2	5	21	26	D-17	26						
			27	27.45	6	8	15	23	UD-12	23						
									D-18							
			28.5	28.95	5	4	21	25	UD-13	25						
									D-19							

RL WITH REPECT TO SUB.

LEGEND FOR BORELOG AND GRAIN SIZE DISTRIBUTION (BASED ON 'ASTM CLASSIFICATION D-2487')

PLASTIC CLAY | SILT | PEAT/HIGHLY ORGANIC SOIL | UNDISTURBED SAMPLE

MICACEOUS FINE SAND/SILT | SAND | SHALE | DISTURBED SAMPLE

N- BLOWS DELIVERED PER FOOT BY A 140 LB HAMMER FALLING 30 INCHES

MAX^{m} PENETRATION= 50 BLOWS. AFTER 50 BLOWS PENETRATION DEPTH WERE MEASURED 50/X = BLOWS 50, "X" MM PENETRATION left of 300 mm

01	BRRL PROSOIL	BOREHOLE NO: BH_BRRL_N_14	Longitude: E 91°6'31.47	Latitude: N 23°3'7.42
		Name of Project: Engineering Soil Mapping of Meghna as per Soil Classification of road master plan (Dhaka, Comilla, Narayangonj, Munshigonj).		
CLIENT : BRRL.		Type of Drill Rig: Mechanical Rotary	Borehole Depth: 51.0m	RL: 2.12m below from Road Level.
Location: Vodogaon, Amisha para, Sonaimuri, Noakhali.		Commencement Date: 06-06-2016	Completed Date: 10-06-2016	G. W. T : 1.83m Below from E. G. L.

THICKNESS	SOIL STRATA	BORE LOG	SAMPLING From (m)	SAMPLING To (m)	SPT-N Value Blow/300mm N1=150 (mm)	N2=300 (mm)	N3=450 (mm)	N=N2+N3	DISTURBED SAMPLE	DEPTH (m) SPT VALUES (BLOWS / 300mm)	% Sand	% Passing 200	% LL	% PI	SP Gravity	% W_n
21.0m	Gray 7.5YR (5/1), dense to very dense, Low Plasticity Silt with Sand, ML		28.5	28.95	5	4	21	25	D-19	25						
			30	30.45	4	4	31	35	D-20	35						
			31.5	31.95	1	10	17	27	UD-14	27						
									D-21							
			33	33.45	4	5	28	33	UD-15	33						
									D-22							
			34.5	34.95	5	12	22	34	D-23	34						
			36	36.45	0	11	25	36	UD-16	36						
									D-24							
			37.5	37.95	6	5	32	37	D-25	37						
			39	39.45	7	6	34	40	UD-17	40						
									D-26							
			40.5	40.95	8	16	26	42	D-27	42						
16.95m	Gray 7.5YR (5/1), very dense, Silty sand, SM.		42	42.45	8	7	37	44	D-28	44						
			43.5	43.95	9	9	38	47	UD-18	47						
									D-29							
			45	45.45	11	11	39	50	D-30	50						
			46.5	46.95	8	8	42	50	D-31	50						
			48	48.45	3	3	47	50	UD-19	50						
									D-32							
			49.5	49.95	12	12	38	50	D-33	50						
			51	51.45	14	14	36	50	D-34	50						

RL WITH REPECT TO SUB.

LEGEND FOR BORELOG AND GRAIN SIZE DISTRIBUTION (BASED ON 'ASTM CLASSIFICATION D-2487')

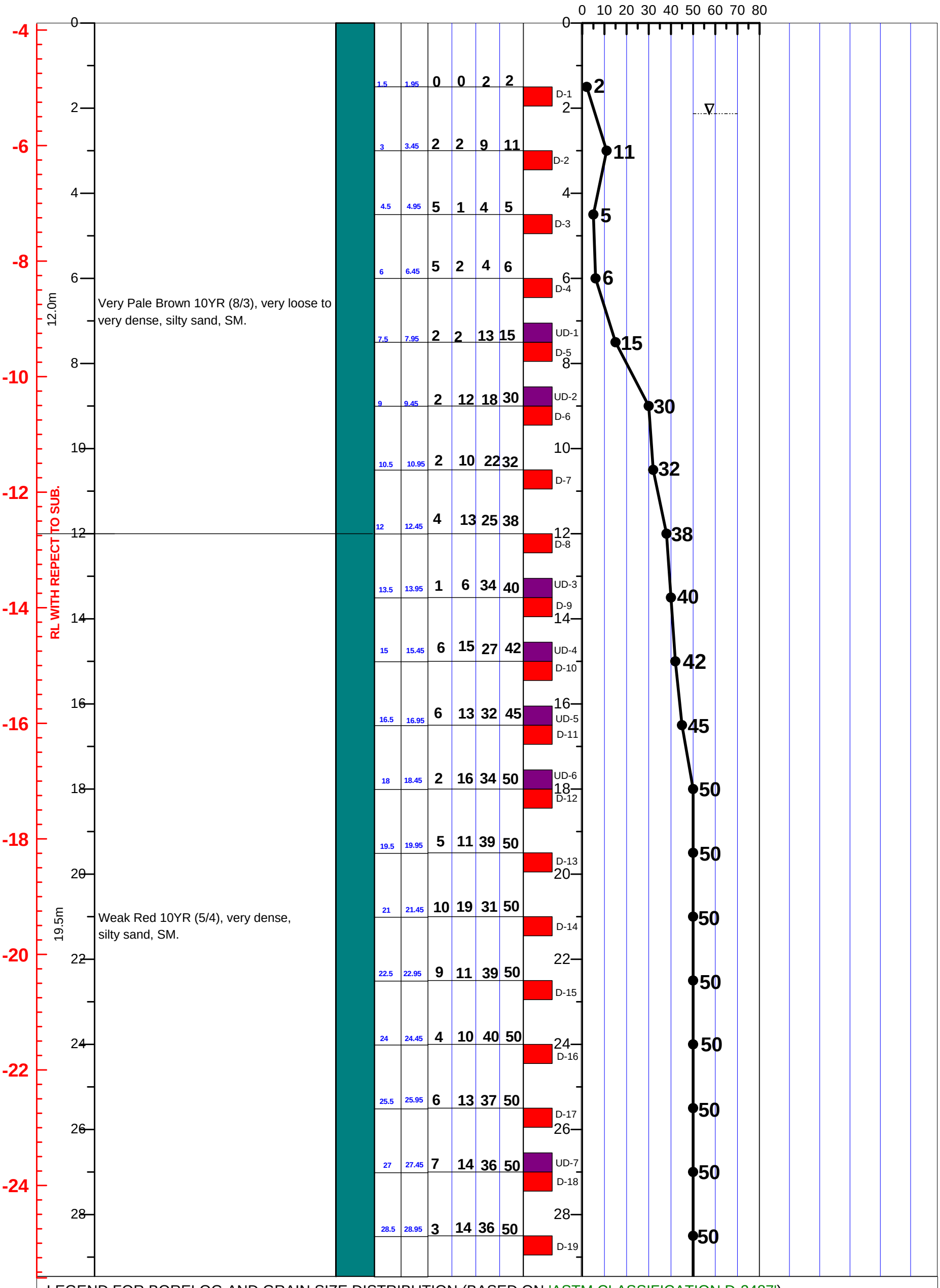
PLASTIC CLAY | SILT | PEAT/HIGHLY ORGANIC SOIL | UNDISTURBED SAMPLE

MICACEOUS FINE SAND/SILT | SAND | SHALE | DISTURBED SAMPLE

N- BLOWS DELIVERED PER FOOT BY A 140 LB HAMMER FALLING 30 INCHES

MAX^{m} PENETRATION= 50 BLOWS. AFTER 50 BLOWS PENETRATION DEPTH WERE MEASURED 50/X = BLOWS 50, "X" MM PENETRATION left of 300 mm

A2	PROSOIL	BOREHOLE NO: BH_BRRL_N-15	Longitude: E 90°58'34.01"	Latitude: N 23°3'41.01"
		Name of Project: Engineering Soil mapping of Noakhali & Duptila as per soil classification of Road Master Plan Noakhali, Feni, Bhola, Chittagong, Dohazari & Cox's Bazar.		
CLIENT : BRRL.		Type of Drill Rig: Mechanical Rotary	Borehole Depth: 51.0m	RL: 3.9m below from Road Level.
Location: Sibpur, Abirpara, Chatkhali, Noakhali.		Commencement Date: 06-06-2016	Completed Date: 10-06-2016	G. W. T : 2.13m Below from E. G. L.



LEGEND FOR BORELOG AND GRAIN SIZE DISTRIBUTION (BASED ON 'ASTM CLASSIFICATION D-2487')

PLASTIC CLAY	SILT	PEAT/HIGHLY ORGANIC SOIL	UNDISTURBED SAMPLE
MICACEOUS FINE SAND/SILT	SAND	SHALE	DISTURBED SAMPLE

N- BLOWS DELIVERED PER FOOT BY A 140 LB HAMMER FALLING 30 INCHES

MAX^{m} PENETRATION= 50 BLOWS. AFTER 50 BLOWS PENETRATION DEPTH WERE MEASURED 50/X = BLOWS 50, "X" MM PENETRATION left of 300 mm

01	BRRL PROSOIL	BOREHOLE NO: BH_BRRL_N-15	Longitude: E 90°58'34.01"	Latitude: N 23°3'41.01"
		Name of Project: Engineering Soil mapping of Noakhali & Duptila as per soil classification of Road Master Plan Noakhali, Feni, Bhola, Chittagong, Dohazari & Cox's Bazar.		
CLIENT : BRRL.		Type of Drill Rig: Mechanical Rotary	Borehole Depth: 51.0m	RL: 3.9m below from Road Level.
Location: Sibpur, Abirpara, Chatkhali, Noakhali.		Commencement Date: 06-06-2016	Completed Date: 10-06-2016	G. W. T : 2.13m Below from E. G. L.

THICKNESS	SOIL STRATA	BORE LOG	SAMPLING From (m)	SAMPLING To (m)	SPT-N Value Blow/300mm N1=150 (mm)	N2=300 (mm)	N3=450 (mm)	N=N2+N3	DISTURBED SAMPLE	DEPTH (m) SPT VALUES (BLOWS / 300mm)	% Sand	% Passing 200	% LL	% PI	SP Gravity	% W_n
19.5m	Weak Red 10YR (5/4), very dense, silty sand, SM.		28.5	28.95	4	11	39	50	D-19	28						
			30	30.45	5	14	36	50	D-20	30						
			31.5	31.95	9	12	38	50	UD-8							
									D-21	32						
			33	33.45	2	17	33	50	D-22							
			34.5	34.95	7	17	33	50	D-23	34						
			36	36.45	5	14	36	50	D-24	36						
19.5m	Gray 10YR (6/1), very dense, silty sand, SM.		37.5	37.95	4	15	35	50	UD-9							
									D-25	38						
			39	39.45	5	14	36	50	UD-10							
									D-26							
			40.5	40.95	6	19	31	50	D-27	40						
			42	42.45	9	20	30	50	D-28	42						
			43.5	43.95	4	17	33	50	D-29							
			45	45.45	15	15	35	50	UD-11	44						
									D-30							
			46.5	46.95	16	16	34	50	UD-12	46						
									D-31							
			48	48.45	15	15	35	50	D-32	48						
			49.5	49.95	14	14	33	50	D-33	50						
			51	51.45	15	15	35	50	UD-13							
									D-34							

RL WITH REPECT TO SUB.

-24, -26, -28, -30, -32, -34, -36, -38, -40, -42, -44, -46, -48, -50, -52

0 20 40 60 80 100

50 50 50 50 50 50 50 50 50

LEGEND FOR BORELOG AND GRAIN SIZE DISTRIBUTION (BASED ON 'ASTM CLASSIFICATION D-2487')

PLASTIC CLAY | SILT | PEAT/HIGHLY ORGANIC SOIL | UNDISTURBED SAMPLE

MICACEOUS FINE SAND/SILT | SAND | SHALE | DISTURBED SAMPLE

N- BLOWS DELIVERED PER FOOT BY A 140 LB HAMMER FALLING 30 INCHES

MAX^{m} PENETRATION= 50 BLOWS. AFTER 50 BLOWS PENETRATION DEPTH WERE MEASURED 50/X = BLOWS 50, "X" MM PENETRATION left of 300 mm